

SPIRIT RIVER, AB, Canada

WMO: 715710

Lat: 55.695N

Lon: 119.230W

Elev: 1015

StdP: 89.71

Time Zone: -7.00 (NAM)

Period: 99-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-28.9	-26.5	-32.1	0.2	-28.9	-29.3	0.3	-26.5	10.0	-2.7	9.3	-2.4	2.9	310	0.718

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.2	25.1	15.1	23.1	14.3	21.4	13.6	16.5	22.7	15.3	21.1	14.3	19.7	3.2	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.1	11.4	18.2	13.0	10.6	16.9	12.0	9.9	16.3	49.9	22.9	46.4	21.3	43.5	19.7	20.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.7	8.0	7.3	-31.5	28.7	3.5	2.2	-34.0	30.3	-36.0	31.5	-38.0	32.7	-40.5	34.3		
(5)			-31.5	18.4	3.5	1.1	-34.0	19.2	-36.1	19.9	-38.0	20.5	-40.5	21.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	2.6	-8.6	-7.2	-5.0	2.5	8.6	13.0	15.1	13.9	9.5	2.9	-4.9	-8.8	(6)
(7)		DBStd	10.90	9.40	7.98	8.00	5.28	4.86	3.41	3.41	3.90	4.63	5.37	7.46	8.58	(7)
(8)		HDD10.0	3176	577	482	464	228	82	11	3	11	62	228	447	582	(8)
(9)		HDD18.3	5749	836	715	722	473	301	163	110	147	266	479	697	841	(9)
(10)		CDD10.0	491	0	0	0	4	40	99	162	130	48	7	0	0	(10)
(11)		CDD18.3	22	0	0	0	0	1	2	10	8	2	0	0	0	(11)
(12)		CDH23.3	155	0	0	0	1	4	16	73	50	11	0	0	0	(12)
(13)		CDH26.7	23	0	0	0	0	0	1	13	8	1	0	0	0	(13)
(14)	Wind	WSAvg	4.0	4.4	4.2	3.9	3.9	3.8	3.6	3.7	3.7	4.2	4.2	4.3	4.5	(14)
(15)	Precipitation	PrecAvg	476	34	21	26	26	44	72	73	52	39	33	34	27	(15)
(16)		PrecMax	684	91	49	47	50	89	144	132	135	110	93	89	106	(16)
(17)		PrecMin	308	14	2	12	7	8	21	24	12	9	8	8	2	(17)
(18)		PrecStd	91	18	11	11	11	22	36	32	31	23	19	18	19	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.7	8.2	10.4	18.7	23.9	25.2	28.4	27.6	25.1	19.0	10.7	8.0	(19)
(20)			MCWB	3.1	2.5	4.2	8.2	11.7	14.3	16.9	16.2	14.6	10.5	5.2	3.0	(20)
(21)		2%	DB	6.0	5.6	7.8	14.6	20.5	23.0	25.3	24.4	21.4	14.8	7.9	5.7	(21)
(22)			MCWB	1.7	1.5	2.5	6.5	10.2	13.1	15.7	15.6	13.4	8.5	3.9	1.8	(22)
(23)		5%	DB	4.0	4.0	6.1	11.9	18.1	21.1	23.1	22.1	18.6	11.9	5.6	3.9	(23)
(24)			MCWB	1.2	0.3	1.5	4.9	9.1	12.6	15.0	14.6	12.0	7.0	2.3	1.1	(24)
(25)		10%	DB	2.4	2.4	4.2	9.7	16.2	19.2	21.2	20.1	16.2	9.8	3.6	2.0	(25)
(26)			MCWB	0.1	-0.6	0.3	3.6	8.2	11.7	14.1	13.5	10.9	5.6	1.1	-0.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.9	3.4	4.9	8.8	12.7	16.2	18.2	17.9	15.9	11.3	6.1	3.9	(27)
(28)			MCDB	7.3	6.7	9.7	17.3	20.4	21.9	24.6	24.2	22.2	17.8	9.4	6.7	(28)
(29)		2%	WB	2.5	1.9	3.1	6.9	11.1	14.4	16.8	16.2	14.0	9.1	4.3	2.4	(29)
(30)			MCDB	5.5	5.1	7.0	14.0	18.1	20.6	23.5	22.2	19.6	13.7	7.2	5.0	(30)
(31)		5%	WB	1.4	0.6	1.7	5.3	10.0	13.3	15.5	15.2	12.5	7.5	2.9	1.2	(31)
(32)			MCDB	3.8	3.6	5.5	10.8	16.7	19.4	21.5	20.6	17.7	11.7	5.4	3.8	(32)
(33)		10%	WB	0.2	-0.5	0.6	4.0	8.8	12.3	14.5	14.1	11.1	5.9	1.4	-0.2	(33)
(34)			MCDB	2.1	2.1	3.9	9.1	14.7	17.9	20.0	18.8	15.3	9.3	3.4	2.2	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.0	7.5	8.6	8.9	9.6	9.2	9.1	8.5	6.6	6.0	6.6	(35)	
(36)			MCDBR	6.8	7.7	9.5	11.2	11.8	11.4	11.7	11.5	11.4	9.6	6.7	6.4	(36)
(37)			MCWBR	4.9	4.9	5.8	6.2	5.7	5.3	5.5	5.4	5.5	5.7	4.6	4.8	(37)
(38)		5% WB	MCDBR	6.9	7.6	9.2	10.6	11.0	10.2	10.7	10.6	11.0	9.0	7.0	6.6	(38)
(39)			MCWBR	5.5	5.2	6.0	6.1	5.6	5.4	5.6	5.6	5.6	5.7	5.4	5.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.229	0.237	0.285	0.302	0.325	0.327	0.327	0.327	0.292	0.271	0.230	0.214	(40)	
(41)		taud	2.311	2.373	2.304	2.435	2.437	2.443	2.471	2.455	2.568	2.578	2.429	2.265	(41)	
(42)		Ebn at Noon	734	874	890	926	918	917	910	888	885	828	740	661	(42)	
(43)		Edh at Noon	49	68	95	98	104	105	100	95	74	57	43	40	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.72	1.66	3.15	4.70	5.61	6.10	5.93	4.82	3.30	1.76	0.79	0.51	(44)	
(45)		RadStd	0.07	0.13	0.25	0.26	0.38	0.48	0.37	0.43	0.29	0.20	0.07	0.04	(45)	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47) Regional (18 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page